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Supplementary Material for:
Counterfactual Audits Must Precede Conformal Trust for Multimodal Medical
Claim Verifiers

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mous Author <anon.email@domain.com>.

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ference on Machine Learning (ICML). Do not distribute.

Table 1. Training hyperparameters shared across all reported configurations. Per-configuration loss-weight settings are derived from the Methods section of the main paper; full per-configuration YAML is in the released codebase.

Image backbone	BiomedCLIP ViT-B/16 (top 4 of 12 blocks trainable)
Text backbone	RoBERTa-large (top 8 of 24 layers trainable)
Shared dim / fusion layers	$d = 768$, 4 cross-modal layers, 12 heads
Verdict head	2-way softmax on a learned [VERDICT] token
Support-score head	sigmoid scalar for conformal calibration
Grounding head	14×14 patch sigmoid map
Optimizer	AdamW
LR (encoders / heads)	1×10^{-5} / 5×10^{-5}
Weight decay	0.01
Warm-up / schedule	500-step linear warm-up, cosine decay to zero
Batch size / grad accum	32 / 2
Precision	bf16
HO-filter activation threshold	0.7 on $p_{\text{text}}^{\text{correct}}$
HO-filter $\mathcal{L}_{\text{cls}}$ downweight	$0.2 \times$
$\mathcal{L}_{\text{consist}}$ margin	0.2
$\mathcal{L}_{\text{contrast}}$ margin	0.2
$\mathcal{L}_{\text{uncert}}$ MC-dropout rate / samples	0.2 / 5
Random seed	42 (global)

Table 2. Single-component drop ablations on the v5.3 configuration. IMG collapses below the 5 pp evidence-blind threshold when either $\mathcal{L}_{\text{consist}}$ or the HO filter is removed; removing the grounding, contrastive, or uncertainty terms each changes IMG by < 5 pp.

Variant of v5.3	IMG (pp)
Full v5.3	69.24
$-\mathcal{L}_{\text{consist}}$ (load-bearing)	2.13
$-\text{HO filter}$ (load-bearing)	5.85
$-\mathcal{L}_{\text{ground}}$	66.10
$-\mathcal{L}_{\text{contrast}}$	63.28
$-\mathcal{L}_{\text{uncert}}$	73.40

A. Threshold Sensitivity for IMG and ESG

The body of the paper treats a verifier as evidence-blind if $\text{IMG} < 5$ pp or $\text{ESG} < 5$ pp. We chose the 5 pp threshold by sweeping $\{3, 5, 7, 10\}$ pp on the validation split and selecting the operating point that maximally separated the v5.0 / v5.1 baselines (whose IMG is ~ 2 pp) from the v5.2 / v5.3 / v6.0 configurations (whose IMG is ≥ 60 pp). The qualitative finding (“the highest-validation-accuracy configuration is evidence-blind”) is robust across all four thresholds tested. ESG is informative on this benchmark only as a coarse pass/fail signal because every trained configuration sits in the $[18.56, 21.13]$ pp band; this is consistent with the dataset-dependence of partial-input gaps reported by ?.

B. Training Hyperparameters

The full composite loss is

$$\mathcal{L} = \lambda_{\text{cls}} \mathcal{L}_{\text{cls}} + \lambda_{\text{ground}} \mathcal{L}_{\text{ground}} + \lambda_{\text{consist}} \mathcal{L}_{\text{consist}} + \lambda_{\text{contrast}} \mathcal{L}_{\text{contrast}} + \lambda_{\text{uncert}} \mathcal{L}_{\text{uncert}}$$

with $\lambda_{\text{cls}} = 1.0$, $\lambda_{\text{ground}} = 0.5$, $\lambda_{\text{consist}} = 0.3$, $\lambda_{\text{contrast}} = 0.3$, $\lambda_{\text{uncert}} = 0.1$.

C. Loss-Drop Ablations

We hold the v5.3 configuration fixed and remove one loss term at a time to identify the load-bearing components. Table 2 reports IMG at the resulting test-time test split.

The two interventions act on disjoint failure modes: the consistency loss penalises verdict-invariance under image masking (a per-example signal), and the HO filter downweights training rows the text-only model already solves

Table 3. Zero-shot detectors on the 500-claim real-RRG subset. RadFlag is a hallucination flagger that emits a binary stability score and is not input-conditioned; IMG/ESG do not apply (n/a).

Detector	Acc	IMG (pp)	ESG (pp)	Blind?
BiomedCLIP (zero-shot)	0.538	8.0	0.0	yes
MedGemma-4B-IT (as verifier)	0.758	14.0	2.0	yes
MAIRA-2 (as verifier)	0.498	32.2	6.6	no
RadFlag (MAIRA-2, $K=6$)	0.664	n/a	n/a	n/a
Ours (v6.0-retrain)	0.911	70.21	20.01	no

(a row-reweighting signal). Both are required.

D. Benchmark Cohort and Silver-Label Protocol

Training cohort. GroundBench-v6 aggregates three public sources: OpenI (?) (US frontal radiographs with paired CheXbert-format reports), ChestX-Det10 (?) (US radiographs with pixel annotations over ten finding categories), and PadChest-GR (?) (bilingual Spanish/English studies with per-sentence radiologist bounding boxes). Patient-stratified splits at 70/10/10/10 via MD5-hash bucketing on the patient identifier yield 28,361 training, 4,046 validation, 4,046 calibration, and 4,132 test claims after filtering for resolved sup/con labels.

Real-RRG silver labels. For real-hallucination evaluation we generate findings-section reports from 500 held-out OpenI studies using MAIRA-2 (?), CheXagent-2-3B (?), and MedGemma-4B-IT (?). Each generated report is decomposed into atomic claims; the three generators yield 2,707 atomic claims after decomposition. We silver-label each claim with a three-LLM ensemble (a MIMIC-fine-tuned grader plus two MIMIC-free graders for leakage decoupling (?)) and report two-of-three majority labels. 1,599 claims resolve to sup, 335 to con, and 773 to uncertain.

E. Field Audit: Per-Detector Numbers

The combination of axes matters: a verifier can fail one axis without failing the other, and aggregate accuracy alone does not predict the joint outcome.

F. Inverted Conformal Selection Procedure

We wrap each trained verifier in an inverted conformal Benjamini–Hochberg selection procedure calibrated against the support-score distribution of con claims on a held-out 4,046-claim calibration split. The procedure follows the conformal-factuality template of ? and the FDR-controlled selection template of ?. At target $\alpha = 0.10$, only v5.3-contrast produces a non-empty trusted set: $n_{\text{green}} = 985$ claims at achieved FDR = 0.91% and power 39.6%. The v5.0, v5.1, and v5.2 configurations collapse to $n_{\text{green}} = 0$.

The mechanism is joint-distribution-driven rather than marginal: the support-score distributions of v5.0 and v5.1 are not less sharp than v5.3 on the marginal axis, but they fail to separate sup from con along the score-label joint, so the calibration set’s con scores cover the test set densely and Benjamini–Hochberg finds no cutoff at $\alpha = 0.10$. This is the empirical statement of the discussion-section claim that “conformal selection on an evidence-blind verifier has nothing to certify.”

G. Reproducibility Checklist

Met: (i) Training data sources are public (OpenI, ChestX-Det10, PadChest-GR); MIMIC-CXR is not used. (ii) Patient-stratified 70/10/10/10 splits via MD5-hash bucketing on patient identifier. (iii) Hyperparameters in Appendix B; per-configuration YAML in the released codebase. (iv) Modal H100 80 GB; bf16 mixed precision.

Deferred to extended version: model and code release URLs (anonymized for double-blind review, filled at camera-ready); multi-seed replication; PadChest-GR claim-extraction for full three-site training.